

Signed Pro/Solid All-Prime Boundary Object

Autonomous discussion Pass 104 assembles the finite signed conductor system into the all-prime pro/solid boundary package.

For $\sigma \in \{\pm 1\}$, the finite signed conductor classes

$$b_N^\sigma = \sigma \bmod N$$

are compatible under conductor reduction. Their inverse limit is the diagonal integer

$$\{b_N^\sigma\}_N = \sigma \in \widehat{\mathbb{Z}}.$$

Since $\sigma = \pm 1$ lies in the diagonal copy of $\mathbb{Z}$, its image in

$$\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}$$

is zero. Thus the two signs do not define two distinct points of ϵ , and the orientation double cover does not survive as a nontrivial point-cover or torsor over the all-prime boundary group.

The sign survives as the $\mathbb{Z}/2$ local-system action on the boundary/Yoneda line. For

$$\delta_\epsilon = [0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0],$$

the action is

$$\delta_\epsilon \mapsto \sigma \delta_\epsilon.$$

Under the shifted-dual identification

$$D\epsilon \simeq \mathbb{Q}[-1],$$

this is the same sign action on the shifted boundary generator.

The minimal sign-preserving all-prime package is:

- the oriented-support action groupoid of pairs (S, c) ;
- the finite-conductor pro-system of signed CRT-isomorphism complexes;
- a $B\mathbb{Z}/2$ local system on the boundary/Yoneda line.

This package records support, conductor, and sign, but it does not create a degree-0 Weyl/Fourier map $\epsilon \rightarrow \mathbb{Q}$.

The next task is to compare this signed pro-boundary stack with support projections and zero-extension spans, then isolate the exact descent/colimit statement for all-prime primitive orientations.

The checker `artifacts/reports/pass104-signed-pro-solid-boundary-object-check.json` verifies conductor compatibility of the finite sign system, identifies the pro-limit as the diagonal integer σ , verifies that both signs vanish as points of ϵ , records survival as a $\mathbb{Z}/2$ boundary action, and checks that the minimal package does not create a degree-0 Weyl map.